

Roni Altshuler

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Summary

Biomolecular Engineer and Bioinformatician exploring how immune and metabolic pathways shape health and disease. Experienced in spatial transcriptomics, CRISPR/Cas9, and cancer genomics, with a focus on translating discovery into therapeutic impact.

Education

Technion – Israel Institute of Technology

2025 – Present

PhD., Biology

Graduate Researcher in the Ron-Harel Lab (Advisor: Dr. Noga Ron-Harel) - focus on ImmunoMetabolism & Aging

University of California, Santa Cruz

2023 – 2024

Master of Science in Biomolecular Engineering and Bioinformatics

Thesis: Spatial Transcriptomic Analysis of Cell Type Distribution and Gene Expression Patterns in a Human Breast Cancer Model

Relevant Coursework: Applied RNA Bioinformatics, Machine Learning, Stem Cell Biology, Entrepreneurship in Biotech

University of California, Santa Cruz

2020 – 2023

Bachelor of Science in Biomolecular Engineering and Bioinformatics with Honors

Dean's Honors List: 2021, 2022, 2023

Relevant Coursework: Bioinformatics Models and Algorithms, Computational Genomics, Data Structures and Algorithms, Database Management Systems, Applied Bayesian Statistics, Python Programming, C Programming, Organic Chemistry, Biochemistry, Molecular Biology

Experience

Research Associate II | Chan Zuckerberg Biohub SF

2024 – 2025

- Designed and implemented a custom Cellpose-based zebrafish cell segmentation model.
 - Developed methods to generate training/testing field-of-view datasets from whole-embryo MERFISH images, including a novel strategy to select training data using **Shannon's entropy**.
 - Increased model F1 score from **53%** to **85%**, significantly enhancing segmentation accuracy to support upstream data analysis processes.
- Contributed to the **Tabula Sapiens Rosetta Donor** project, integrating isoform-level information with traditional single-cell gene expression analyses to add a new dimension to the consortium's data resources.
- Performed high-throughput sequencing for internal and external investigators, including QC workflows (TapeStation, qPCR), sequencing (MiSeq i100, NextSeq 2000, and NovaSeq X), and post-run analysis (demultiplexing reports, AWS-based data tokenization and delivery).

Master's Research | University of California, Santa Cruz Genomics Institute

2023 – 2024

- Developed a custom Cellpose2.0 model for spatial transcriptomic data analysis, achieving precision and recall metrics of **84%** and **90%**.
- Identified critical gene expression patterns through spatial differential analysis, revealing tumor microenvironment insights.
- Visualized spatial distributions and performed clustering analyses using Python tools like UMAP.

CRISPR-X Intern | CRISPR Therapeutics

2023

- Led a project on 3'-overhang dsDNA integration, boosting CRISPR/Cas9 knock-in efficiency (validated via NGS).
- Optimized T cell editing by integrating promoterless GFP transgenes and analyzing results with flow cytometry.

Autoimmune Intern | CRISPR Therapeutics

2022

- Enhanced CAR-T cell performance by analyzing costimulatory domain variations and improving targeting efficiency.
- Evaluated cancer cell targeting outcomes using advanced flow cytometry and ddPCR techniques.

Technical Skills

Programming & Tools: Python, HPC, Linux/Unix, Git, R, Jupyter, CellPose, AlphaFold, Matplotlib, Squidpy, Scanpy, Seaborn, Pandas, AWS

Laboratory Techniques: Next Generation Sequencing, CRISPR/Cas9 Editing, PCR (qPCR, ddPCR), Flow Cytometry, Gel Electrophoresis, Tissue Culture

General: SnapGene, GraphPad Prism, FlowJo, Microsoft Office, Google Suite

Presentations

Poster Presentations

1. “Consecutive Sections MERFISH Reveals Spatially Resolved Gene Expression Dynamics in 24hpf Zebrafish Embryos”
*Amanda Seng, **Roni Altshuler**, Shruthi VijayKumar, Andy Zhou, Xiang Zhao, Merlin Lange, Loic A. Royer, Norma Neff.* Biohub SF Inter-lab Confab. May 2025. San Francisco, USA.
2. “Emerging Transcriptomics at CZ Biohub SF”
*Ritwicq Arjyal, Amanda Seng, Andy Zhou, Thomas Gottilla, **Roni Altshuler**, Norma Neff.* Biohub SF Inter-lab Confab. May 2025. San Francisco, USA.
3. “Investigating The Cellular Response to External Perturbations At a Molecular Level Using High-Throughput Screens”
*Leah C. Dorman, Karen Cheng, Chad Liu, Duo Peng, Cassidy Jones, Madhurya Sekhar, Ritwicq Arjyal, Thomas Gottilla, **Roni Altshuler**, Norma Neff, Carolina Arias, Manu Leonetti, Yasin Senbabaoglu.* Biohub SF Inter-lab Confab. May 2025. San Francisco, USA.
4. “Emerging Spatiotemporal Transcriptomics at CZ Biohub SF”
*Ritwicq Arjyal, Amanda Seng, Andy Zhou, Thomas Gottilla, **Roni Altshuler**, Norma Neff.* Biohub SF Inter-lab Confab. October 2024. San Francisco, USA.
5. “Spatially Dependent Differential Analysis of T-Cells in Hepatocellular Carcinoma using Spatial Transcriptomics.”
*Michael Borja, Nick Keener, Cecilia Cisar, **Roni Altshuler**, Uyen Nguyen, Benedict Paten.* UCSC Poster Session. March 2024. Santa Cruz, USA.